

Physical-Only Proprioceptive Recognition of Household Objects with a LEAP Hand: A Four-Class Pilot Toward Twenty Classes

G. Sai Pranav, Harish J. Palanthandalam-Madapusi

Abstract

Robotic hands commonly use vision or dedicated tactile sensors to recognize objects. This study examines whether a physical LEAP Hand can identify household items using only the joint angles reached after controlled grasps. Motor-current thresholds stop an entire finger when resistance is detected. Final angles from an ordered sequence of grips form the input to a compact neural classifier; all training measurements come from the physical hand. The planned study will enroll 20 household objects. The available household pilot contains a ball, bottle, bowl, and computer mouse, with three real trials per class. Its earlier single-grip neural model correctly classified 11 of 12 held-out trials in three-fold validation (91.7%). Ten predictions passed its confidence and distance rules, and all ten were correct. A nearest-trial baseline also scored 11/12. The current software supports three sequential grips, but no three-grip or 20-class performance has yet been measured. These pilot results demonstrate recognition among the four recorded items; the broader accuracy and robustness questions remain open.

Keywords: proprioception, active perception, LEAP Hand, household object recognition, neural network, physical grasping

1. Introduction

A robot can learn about an object by interacting with it. For a dexterous hand, the joint positions reached while closing around an object depend on the surfaces that interrupt finger motion. These positions may therefore contain information about the object even without an external camera or dedicated tactile skin. This use of the hand’s internal state is known as

proprioceptive perception. The LEAP Hand is a dexterous, anthropomorphic hand developed for robot learning and manipulation (Shaw et al., 2023).

Previous work has shown that joint information can support object recognition. Vásquez and Perdereau (2017) developed proprioceptive shape signatures from robotic-hand joint angles and studied shape identification across object sizes and poses. Sintov and Meir (2023) used grasp information for kinesthetic object recognition without dedicated tactile sensors. More recently, Kutsuzawa et al. (2024) investigated learning-based shape and stiffness estimation from joint angles and velocities in a multi-fingered hand. These studies motivate a physical-hand experiment using a simple observation: the final joint angles of a repeatable grip sequence.

The main research question is whether final angles from the physical LEAP Hand can distinguish 20 enrolled household objects. A second question is whether combining repeated grip sequences improves decisions that are uncertain after one sequence. The model predicts object identity, rather than object size. Unless multiple examples of a household category are collected, each label refers to one physical item; recognition of another item of the same kind is a separate generalisation question.

The present implementation provides repeatable grips, a resumable physical-data collection procedure, a neural classifier, a nearest-trial comparison method, and an interface for running the hand and viewing the results. The available physical trials permit a preliminary four-object evaluation. The planned 20-object experiment has not yet been completed, so its performance is not inferred from the pilot.

2. Experimental Setup and Methodology

2.1. *Physical hand and controlled grips*

We used a four-fingered LEAP Hand with 16 actuated joints (Shaw et al., 2023). The recorded joint vector was ordered as index joints 0–3, middle joints 4–7, ring joints 8–11, and thumb joints 12–15. All recorded angles were expressed in degrees. Objects were presented at a consistent position relative to the palm, and the same programmed grip sequence was used for every object. This controlled placement is an experimental condition rather than a property of the model.

For each configured grip, the hand moved to that grip’s starting posture. Its 16 joints then advanced toward specified targets using the per-joint step sizes in the experiment settings. During closure, motor current served as

an indicator of resistance. If a moving joint’s signed current exceeded its configured threshold, all four joints of that finger stopped at their measured positions. The remaining fingers continued until resistance or an angle target stopped them. A grip ended when all four fingers had stopped and was aborted after at most 250 iterations. The hand moved to the next grip’s starting posture before continuing the sequence.

After each grip, the program read its final 16 joint angles, retrying for up to 10 seconds if position reporting was temporarily unavailable. The final vectors were concatenated in grip order. The current settings contain three grips, yielding 48 input angles per physical trial. Earlier pilot measurements used one grip and therefore yielded 16 angles per trial. The current and historical grip settings are not interchangeable, so the earlier pilot accuracy cannot be attributed to the three-grip procedure.

During collection, the final angles and time-series log were saved before the hand returned to its starting posture. Motor currents controlled the movement but were not classifier inputs. No camera images or tactile arrays were used as recognition features.

2.2. Physical dataset and proposed 20-object experiment

Each training record consists of a household-object label and one concatenated final-angle vector,

$$\mathbf{q}_i = [\mathbf{q}_i^{(1)}, \dots, \mathbf{q}_i^{(G)}] \in \mathbb{R}^{16G},$$

where G is the number of configured grips and each $\mathbf{q}_i^{(g)}$ contains the 16 final joint angles from grip g . The data-collection program stores every completed trial with its final angles and time-series log. It also saves a collection plan before recording begins, so interrupted sessions can resume without discarding earlier trials. Additional trials and new classes can be added later. A mistaken trial may be retaken; the previous record is archived after a complete replacement has been checked.

The intended full study contains 20 distinct household objects, with a target of five to ten independent physical trials per object, or 100–200 labeled trials. The selected objects should be graspable within the hand’s configured motion limits and should vary in geometry and size. Their names, dimensions, and placement orientation will be recorded with the dataset. Repeated grasps should include natural variation in reseating the same object. A separate later session should be reserved to test changes in placement and hand

state. These 20-object trials have not yet been collected in the repository examined for this manuscript.

The available household pilot has three trials each of a ball, bottle, bowl, and computer mouse. Both the neural classifier and nearest-trial baseline were evaluated on these 12 measurements. All trials came from physical objects and used the earlier single-grip configuration. No trained 20-class model or three-grip household dataset is currently available.

2.3. Neural network and training

The neural network receives only the $16G$ final joint angles. Each angle is normalized by the start and target angles of its grip and joint:

$$x_{g,j} = \frac{q_{g,j} - a_{g,j}}{b_{g,j} - a_{g,j}},$$

where $a_{g,j}$ and $b_{g,j}$ are the start and target angles for joint j in grip g . A joint with equal start and target values is assigned a normalized value of zero.

The classifier has $16G$ input units, one fully connected 24-unit hidden layer with a hyperbolic tangent activation, and one output for each enrolled object. A softmax operation converts the outputs to class probabilities. For C classes, the model has $384G + 24 + 25C$ trainable parameters. The current three-grip, 20-class design would have 1,676 parameters; the historical single-grip, four-class pilot had 508. The model does not predict object size.

We used full-batch Adam optimization with a learning rate of 0.01, weight decay of 0.001, and a limit of 500 epochs. Each real trial was retained and supplemented with 20 noisy copies during training. Independent zero-mean Gaussian noise with a standard deviation of 0.5° was added to each angle. A second 0.5° Gaussian perturbation was shared by the four joints of each finger within each grip. These artificial copies approximate small measurement and grasp variation; they are not counted as additional physical trials. Samples were weighted inversely to the number of real trials in their class to reduce the effect of unequal class sizes.

2.4. Validation and recognition decisions

The implemented evaluation uses up to three folds, with real trials from every class assigned to different folds before noise is added. Noise augmentation is performed only on each training fold. The validation fold therefore contains measurements from the physical hand, not perturbed copies

of training samples. The best epoch in each fold is selected by validation cross-entropy, with training stopped after 80 epochs without improvement. The final model is then trained on all real trials for the median number of selected epochs.

The class probabilities are adjusted with a temperature selected from the held-out folds. We also compute the distance from a new normalized angle vector to the nearest saved physical training grasp. A prediction is reported as *unknown* when this distance exceeds 1.5 times the 95th percentile of the validation distances. Otherwise, it is reported as *uncertain* when the largest class probability is below 0.75 or the difference between the two largest probabilities is below 0.20. A known class is returned only when these checks pass. These thresholds were chosen using known-class trials; their ability to reject genuinely unseen objects has not yet been evaluated.

If a neural prediction remains uncertain, the program can request another trial of the same object, up to three trials. Each trial runs the configured grip sequence. The class probabilities are averaged across trials, while the nearest-reference distances are combined using their median. The resulting decision is saved with each trial. A separate nearest-trial method assigns the class of the saved physical grasp with the smallest Euclidean distance in degrees. That method supplies a comparison baseline; it does not produce probabilities or an unknown-object decision.

For the completed 20-object study, the primary outcome will be raw class accuracy on held-out physical trials, reported with a confusion matrix and per-class results. We will also report the fraction of predictions accepted by the decision rule, the accuracy among accepted predictions, and the effect of repeated grasps. Trials from later sessions and separate objects excluded from training are needed to evaluate robustness and unknown-object rejection independently. The current folded evaluation does not substitute for those tests.

2.5. Software implementation

The grip and physical-only training programs are part of Ex-ARM (Pranav, 2026). The terminal interface can collect, resume, extend, and retake physical trials, train either model, and run the hand for recognition. The presentation window shows the final decision, the neural probabilities for every enrolled class, and a visualization driven by the measured hand pose. The classifier itself uses only recorded physical angles.

3. Results and Discussion

3.1. Four-object household pilot

The available household dataset contains 12 physical trials: three each for a ball, bottle, bowl, and computer mouse. All four labels refer to the specific items recorded. These measurements used the earlier single-grip, 16-angle configuration. The current three-grip version of the software has not yet been evaluated on these objects.

In three-fold validation, the neural classifier assigned the correct class to 11 of the 12 held-out trials (91.7%). The one error assigned a ball trial to the bottle class. Table 1 gives the per-class counts. The final model was then trained on all 12 trials; this training step is distinct from the held-out evaluation. A nearest-trial baseline also classified 11 of 12 trials correctly in leave-one-trial-out evaluation.

Table 1: Held-out neural predictions for the physical household pilot. Each row contains three real trials. The single error was a ball predicted as a bottle.

Actual item	Ball	Bottle	Bowl	Mouse	Correct
Ball	2	1	0	0	2/3
Bottle	0	3	0	0	3/3
Bowl	0	0	3	0	3/3
Mouse	0	0	0	3	3/3
Total					11/12

When the pilot’s probability, margin, and distance rules were applied, ten of the 12 held-out predictions were accepted as known classes. All ten accepted predictions were correct; two were marked uncertain and none were marked unknown. This 10/10 accepted result is descriptive: the temperature and distance limit were selected using the same folded validation predictions. It is not an independent estimate of the acceptance rule’s accuracy. No item excluded from the four training classes was used to evaluate unknown-object rejection.

3.2. Interpretation and study limitations

The pilot supports a narrow conclusion: final joint angles from a controlled physical grasp distinguished these four recorded items in 11 of 12 held-out trials. It does not show that the classifier can recognize 20 different

household objects. Three trials per class give a limited view of within-item variation, and the validation trials came from the same physical exemplars and collection setting as the training trials. The results do not measure generalization to another bottle, ball, bowl, or mouse, to a later session, or to changed placement.

The three-grip design may provide additional information when one grip leaves two objects in similar joint postures. It also makes a trial longer and may disturb the object while the hand returns to the next starting posture. Those effects must be measured rather than assumed. The four-object pilot results cannot be transferred to the three-grip configuration because its inputs, grip motions, and saved model are different.

For the 20-object study, each class should have several independent placements, and trials should be collected across multiple sessions. In addition to a trial-level split, a later session should be held out until the model and its decision thresholds are fixed. Reporting the confusion matrix, per-class recall, overall raw accuracy, coverage, and accuracy among accepted predictions will show which item pairs are difficult to separate. Objects excluded from all training are needed to evaluate the *unknown* output. Repeated-sequence recognition should be compared with a single sequence on the same held-out cases to determine whether another observation improves the decision.

4. Conclusion

This work develops a physical-only proprioceptive recognition procedure for a LEAP Hand. Motor-current thresholds stop contacted fingers, and a compact neural network predicts object identity from final joint angles. The software now supports an ordered sequence of three grips, giving 48 angle inputs per trial, as well as resumable collection, trial retakes, and repeated recognition attempts.

The available household pilot used the earlier single-grip configuration. Its neural classifier correctly identified 11 of 12 held-out physical trials from a ball, bottle, bowl, and mouse. Ten of 12 predictions passed its acceptance rules, and all ten were correct. These findings establish a promising four-item pilot. Collecting and evaluating the 20-class, three-grip dataset is the next step before claiming accurate recognition across 20 household objects.

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